



ASSIGNMENT 1

AZURE AI EXERCISE

Assignment Details:

Course Name: Machine Learning for Business Professionals

1. **Course Code:** ISYS3466

Lecturer: Lecturer. Thi Xuan May Le

Submission Date: 31 March 2026

Word Counts: 2200

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Assessment Declaration:

I declare that in submitting all work for this assessment I have read, understood and agree to the content and expectations of the Assessment declaration.

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1. Identification of potential problem

1.1. Cell_Phone_Price_Prediction dataset

The business problem for the cell phone dataset is to classify phones into price categories based on their technical specifications such as RAM, battery capacity, screen size, processor-related features and camera capability. In a real business setting, this problem matters for smartphone manufacturers, retailers and e-commerce platforms. A reliable classifier can support new product positioning, competitor benchmarking, faster assortment planning and pricing strategy (Ahmed Mahoto et al. 2021). Instead of relying only on expert judgement, firms can use historical patterns to estimate which specification bundles are associated with low, medium or premium price segments. This improves pricing consistency and reduces the risk of launching products at the wrong value tier (Ahmed Mahoto et al. 2021).

Machine learning provides a data-driven solution because the relationship between technical features and final price category is multivariate. Customers do not respond to RAM, battery power or display resolution in isolation; they respond to a bundle of specifications that collectively signal quality. A supervised learning model can identify these patterns and learn decision boundaries between price classes more consistently than manual rules (Shetty et al. 2022). The output is actionable because it helps decision-makers answer practical questions such as: “If we increase RAM and camera quality but keep battery power constant, does the phone move into a higher segment?” This directly aligns with evidence-based pricing and product portfolio management.

1.2. Wine dataset

The wine dataset’s objective is not to predict a known target for future observations, but to discover natural groupings of wines based on chemical attributes such as alcohol, phenols, flavonoids and proline. In commercial terms, clustering can support product differentiation, inventory segmentation, targeted marketing and quality profiling (Kasem et al. 2024). For example, wineries, distributors or specialty retailers can use cluster profiles to separate wines into chemically distinct groups that may correspond to style, quality perception or production origin.

Machine learning, especially unsupervised learning, is suitable because clustering detects hidden structure in unlabelled data. Chaudhry et al. (2023) stated that even where class labels exist in the historical dataset, clustering remains useful because in real operational settings

managers often do not know the “true” groups in advance. The technique can therefore reveal whether wines naturally separate into meaningful categories, and which variables are most influential in that separation (Chaudhry et al. 2023). This is valuable for market segmentation and product portfolio decisions by transforming complex chemical data into actionable insights.

2. Understandings of data characteristics

2.1 Cell_Phone_Price_Prediction dataset characteristics

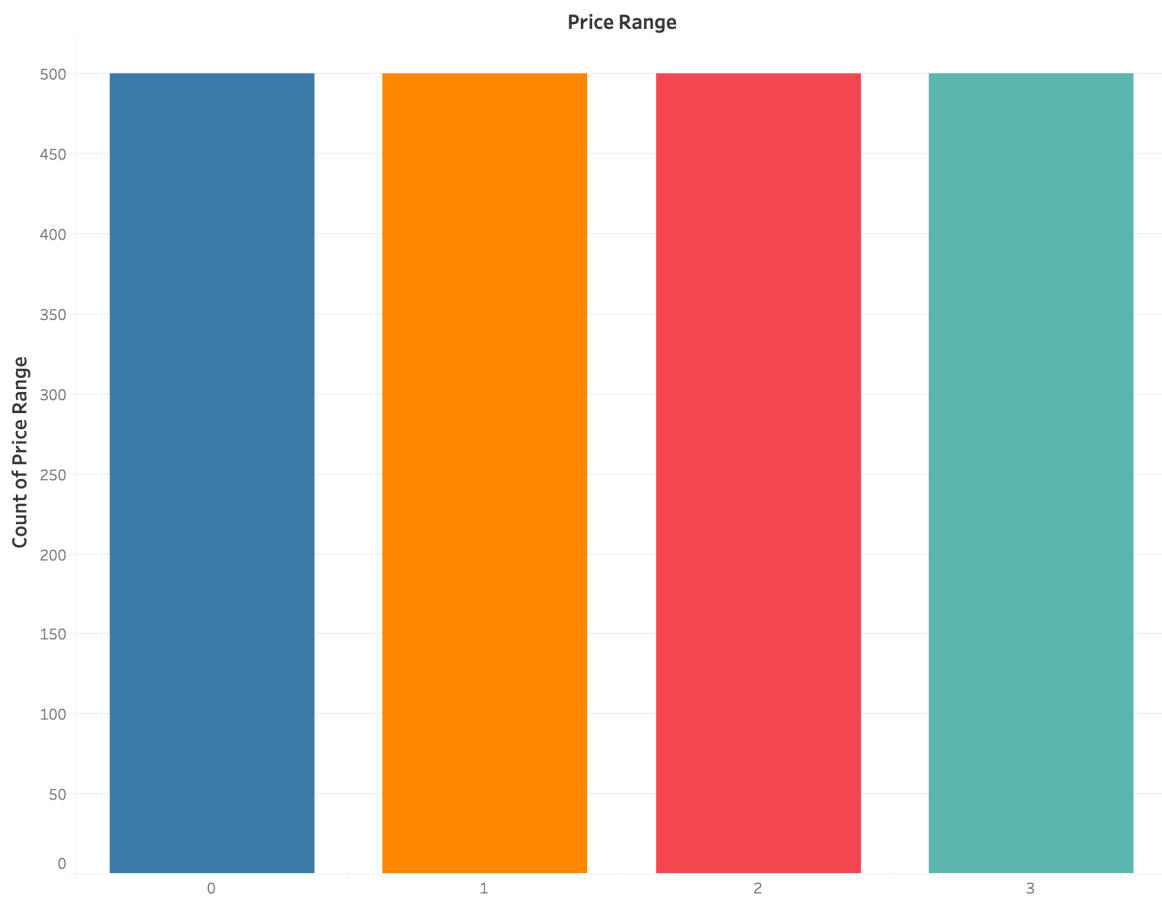


Figure 1: Histogram of price_range distribution.

Feature Names	λ
Battery Power	0.2007
Blue	0.0206
Clock Speed	-0.0059
Dual Sim	0.0168
Fc	0.0214
Four G	0.0148
Int Memory	0.0444
M Dep	0.0009
Mobile Wt	-0.0302
N Cores	0.0043
Pc	0.0336
Px Height	0.1489
Px Width	0.1657
Ram	0.9171
Sc H	0.0230
Sc W	0.0385
Talk Time	0.0216
Three G	0.0236
Touch Screen	-0.0304
Wifi	0.0188

Figure 2: Correlation heatmap of price_range with other attributes.

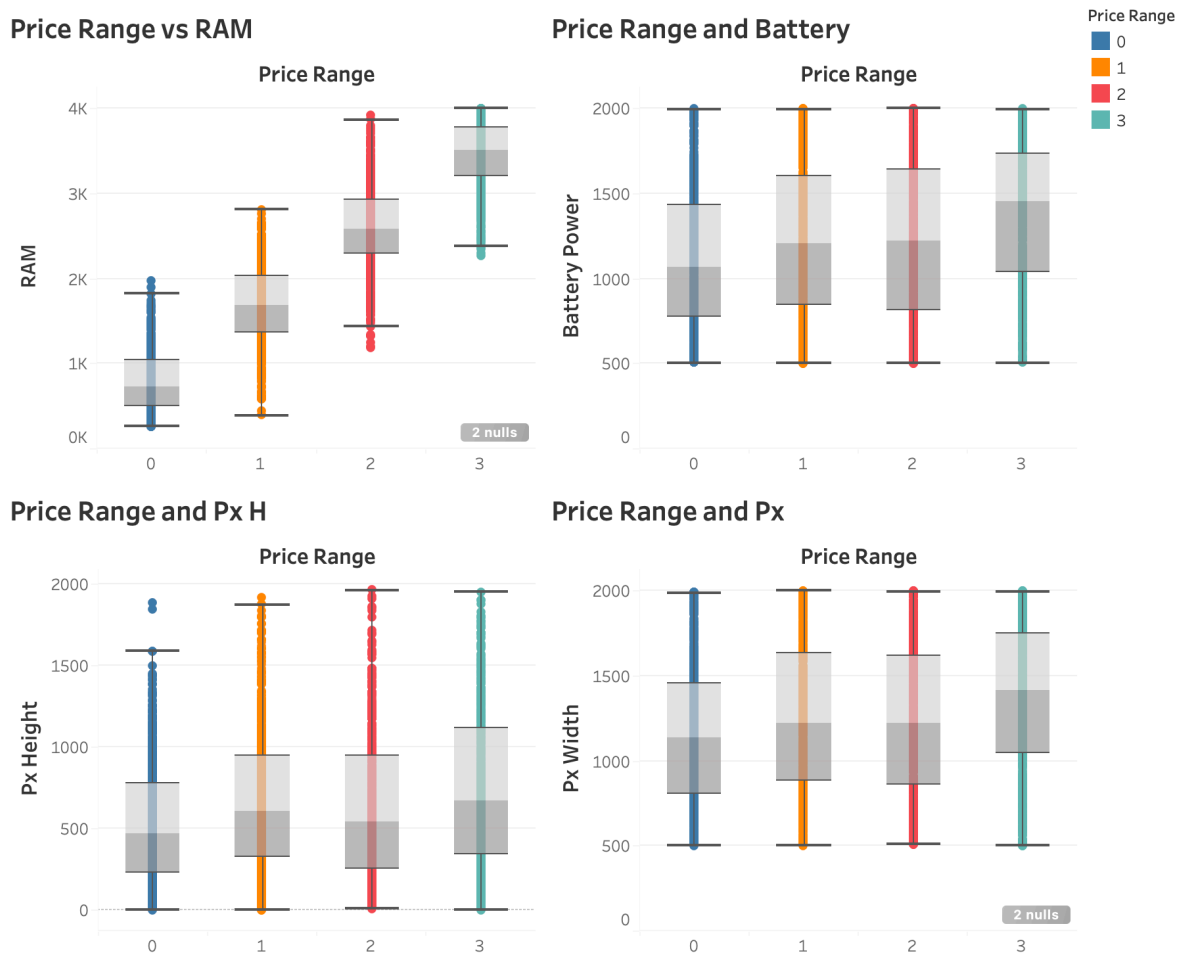


Figure 3: Boxplots of price_range and 4 continuously numerical variables.

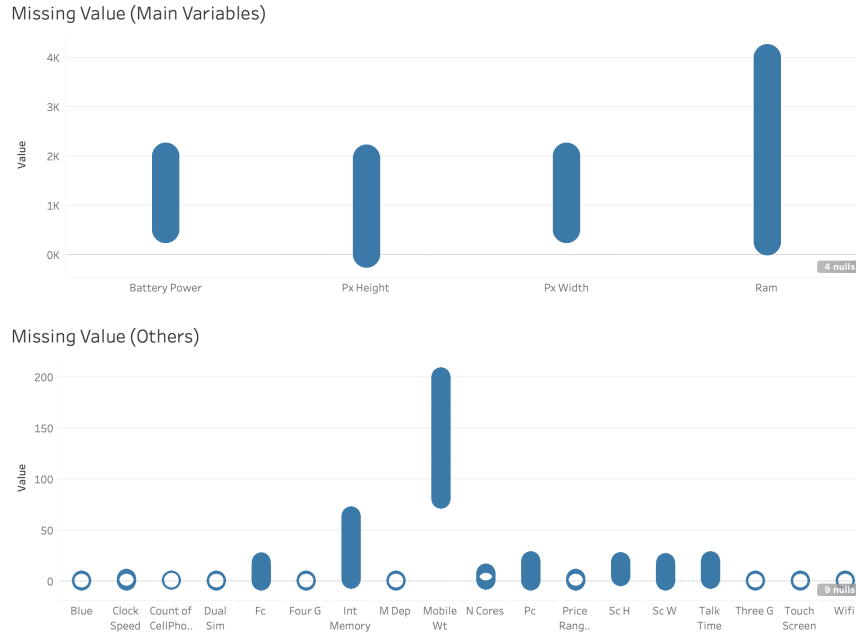


Figure 4: Missing values of Cell_Phone_Price_Prediction training dataset, due to 0 missing values in testing dataset, therefore, no visualisation is indicated.

The datasets are sourced from Kaggle (2024), consisting of a labelled training set with 2,000 observations and an unlabelled testing set of 1,000 observations including an *id* attribute. The training data includes 20 predictor variables and one target variable (*price_range*), enabling supervised classification. Most variables are numerical, including key continuous features such as *ram*, *battery_power*, *px_height*, and *px_width*, along with binary indicators (e.g., *dual_sim*, *wifi*), capturing both technical specifications and product characteristics.

Distribution analysis (Figure 1) shows that *price_range* is evenly distributed across four classes (0–3), indicating a balanced dataset that reduces bias and improves reliability (Dube and Verster 2023). Correlation analysis, commonly used to identify influential features (Kumar and Chong 2018), reveals that RAM has a strong positive relationship with price (~ 0.92), making it the most significant predictor (Figure 2). Other variables, such as battery power (~ 0.2), pixel width (~ 0.17), and pixel height (~ 0.15), show moderate correlations, while most features contribute minimally.

Boxplots (Figure 3) further confirm that RAM has strong discriminative power, with clear separation across price categories, whereas other variables show partial overlap. Data inspection (Figure 4) identifies only 13 missing values (including 4 in key predictors), which are addressed using median imputation due to its robustness and minimal distortion (Joel et al.

2024). Additionally, scale heterogeneity requires normalisation for models such as logistic regression, while tree-based models remain unaffected [1].

From a business perspective, features such as RAM and display quality are key drivers of price segmentation, supporting their importance in predictive modelling.

2.2 Wine dataset characteristics

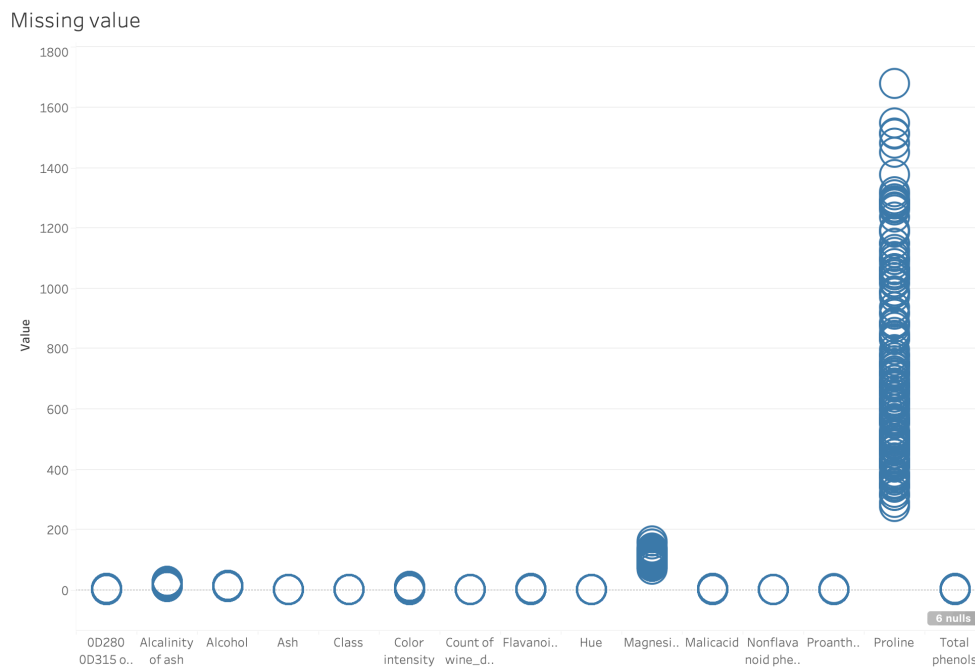
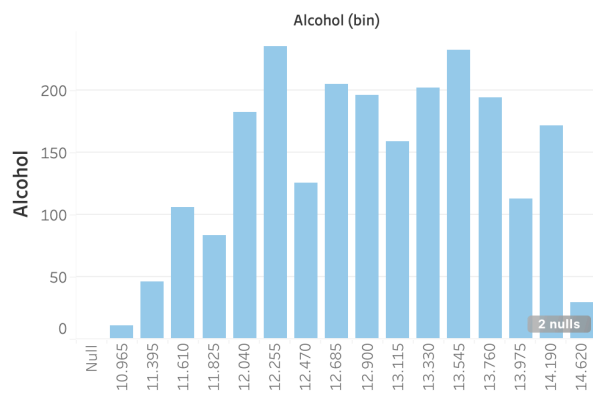
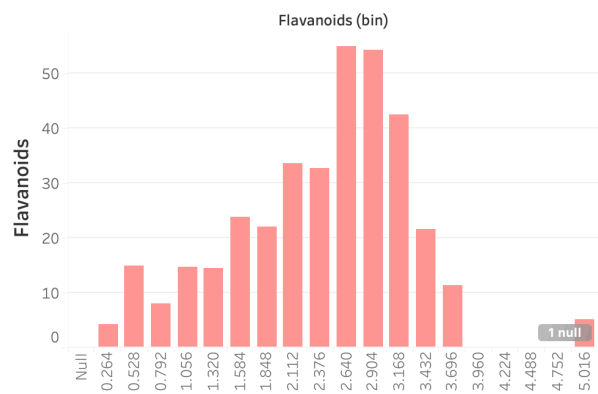


Figure 5: Missing values of Wine dataset.

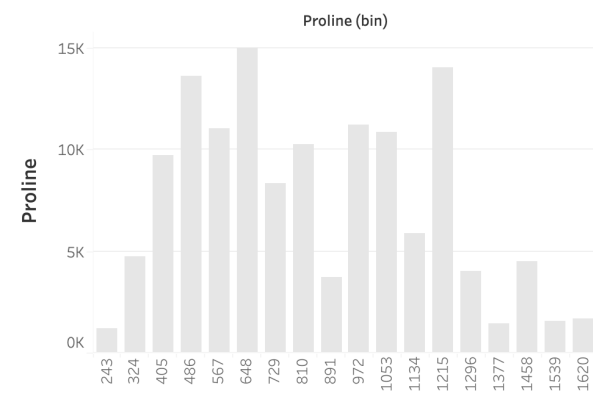
Alcohol Distribution



Flavanoids



Proline



Color Intensity Distribution

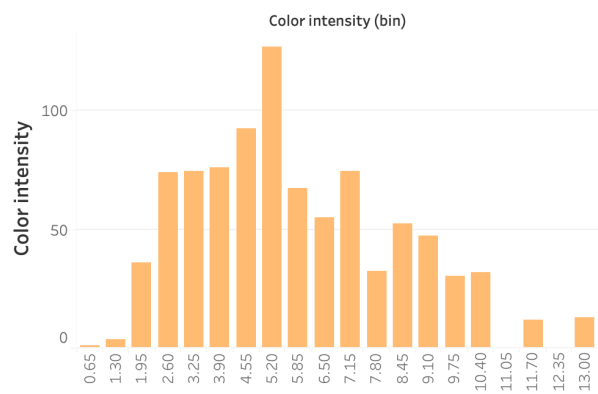


Figure 6: Histograms of Alcohol, Proline, Flavanoids, and Color Intensity.

Feature Name	
0D280 0D315 of diluted w..	0.0747
Alcalinity of ash	-0.3246
Ash	0.1995
Color intensity	0.5468
Flavanoids	0.2253
Hue	-0.0619
Magnesium	0.2704
Malicacid	0.0910
Nonflavanoid phenols	-0.1593
Proanthocyanins	0.1340
Proline	0.6413
Total phenols	0.2812

Figure 7: Correlation heatmap of Alcohol with other attributes.

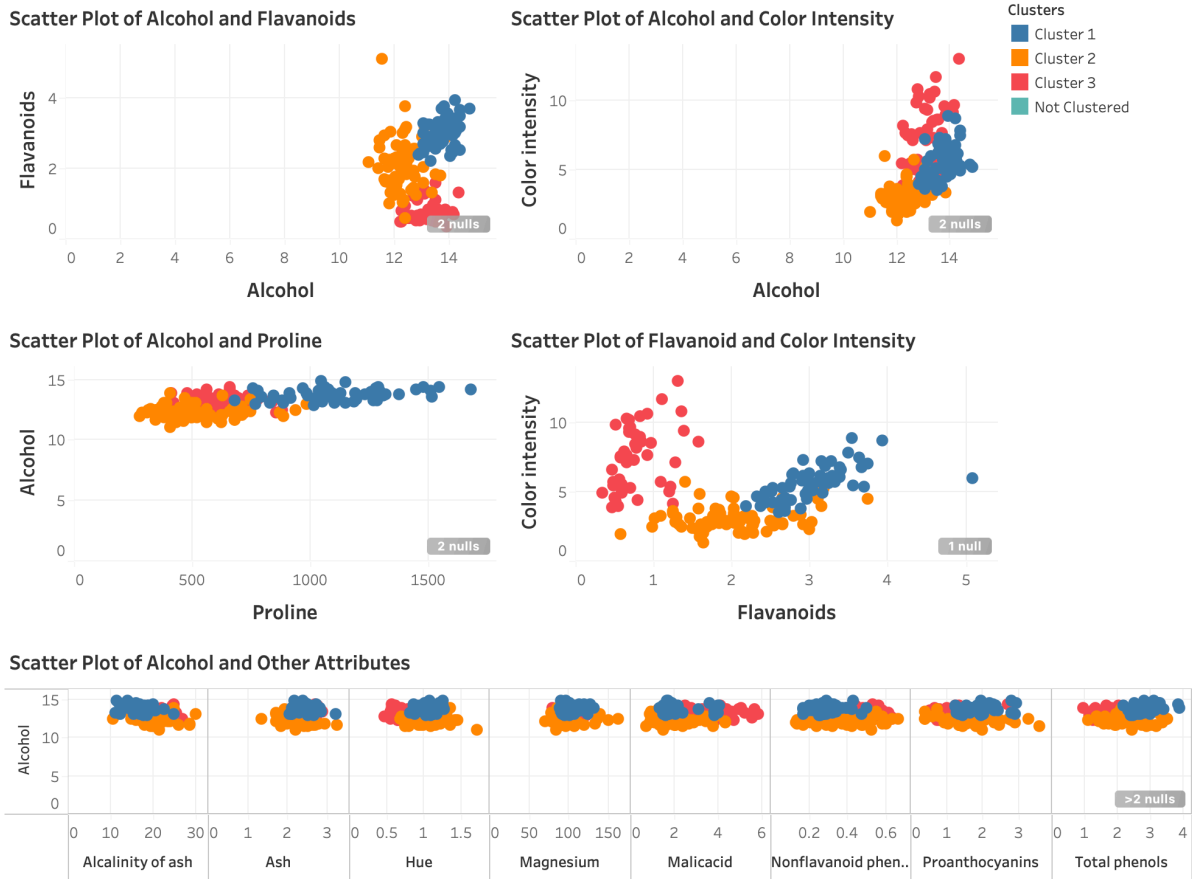


Figure 8: Scatter Plot of Alcohol vs Flavanoids, Proline, Color Intensity, Other Attributes and Flavanoids vs Color Intensity.

The Wine dataset, sourced from the UCI Machine Learning Repository, contains 178 observations and 14 variables, including 13 continuous chemical attributes and a class variable with three categories. Key features include alcohol, flavanoids, proline, color intensity, and magnesium, making it a classic multivariate dataset well-suited for clustering, as distance-based methods perform effectively on numerical data (El Habib Kahla et al. 2024). Data inspection (Figure 5) identifies only 4 missing values, indicating high data quality and minimal need for complex imputation (Joel et al. 2024).

Distribution plots (Figure 6) show variability and mild skewness across key variables. Proline exhibits a wide range with potential outliers, while flavanoids and color intensity are more concentrated, suggesting stronger discriminatory power. Although correlation analysis (Figure 7) provides insight into linear relationships, clustering depends on distance and structure. Variables such as flavanoids remain important due to their strong cluster separation, highlighting the limitation of correlation-based evaluation in unsupervised learning (Rickert et al. 2023).

Scatter plots (*Figure 8*) further confirm clear cluster structure. Strong separation is observed in alcohol vs flavanoids and color intensity, as well as flavanoids vs color intensity, while alcohol vs proline also shows distinct grouping. Other variables exhibit greater overlap and lower discriminative ability.

Based on these patterns, clusters can be meaningfully interpreted: Cluster 1 represents wines with higher alcohol, flavanoids, and proline (premium wines), Cluster 2 reflects moderate attributes (mid-range), and Cluster 3 captures lower values, particularly in flavanoids and proline (lower-quality wines).

3. Data preparation

The data preparation processing is applied in Microsoft Azure AI.

For the Cell_Phone_Price_Prediction dataset, data preparation involves 6 steps.

Step 1: Selecting all columns from the predictors in training dataset (*Figure 9*).



Figure 9: Select Columns in Dataset.

Step 2: Missing values were handled using median imputation across all variables (*Figure 10*), ensuring robustness to outliers while preserving the underlying data distribution. Given the low proportion of missing values, no additional indicator variables were introduced, and columns with entirely missing values were removed to maintain data integrity (Josse et al. 2024).

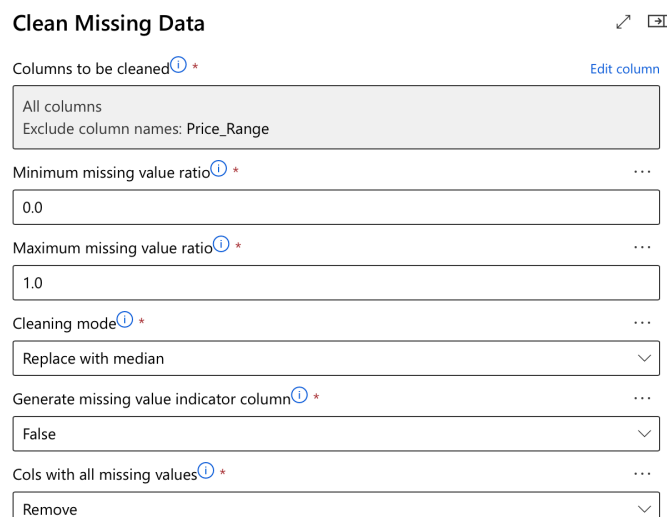
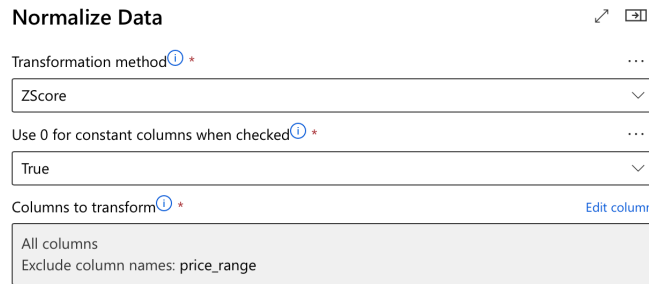


Figure 10: Clean Missing Data.

Step 3: Z-score normalisation was applied to standardise features to a common scale with zero mean and unit variance (Pranolo et al. 2024; *Figure 11*). This is essential due to heterogeneous variable ranges and is particularly important for scale-sensitive models. Compared to min–max scaling, Z-score is more robust to outliers and better preserves the data distribution (Pranolo et al. 2024). The target variable was intentionally excluded, as it represents categorical class labels and should not be transformed to avoid distortion of class boundaries.

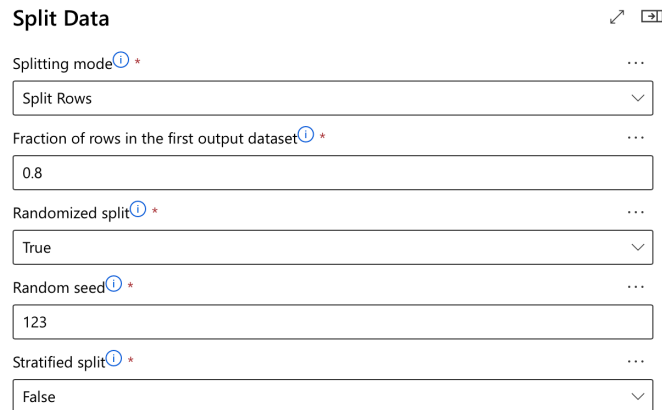


The screenshot shows a 'Normalize Data' configuration window. It includes a title bar with a share icon and a close button. The configuration options are: 'Transformation method' set to 'ZScore', 'Use 0 for constant columns when checked' set to 'True', and 'Columns to transform' set to 'All columns' with 'price_range' excluded. There is an 'Edit column' link next to the columns list.

Normalize Data	
Transformation method	...
ZScore	
Use 0 for constant columns when checked	...
True	
Columns to transform	Edit column
All columns Exclude column names: price_range	

Figure 11: Normalize Data.

Step 4: The dataset was split into 80% training, and 20% validation sets to enable fair model comparison (*Figure 12*). A holdout evaluation approach was used, ensuring an unbiased assessment of model performance on unseen data (Larcher and Barbosa 2022). A fixed random seed was applied to ensure reproducibility and consistency in the data split.



The screenshot shows a 'Split Data' configuration window. It includes a title bar with a share icon and a close button. The configuration options are: 'Splitting mode' set to 'Split Rows', 'Fraction of rows in the first output dataset' set to '0.8', 'Randomized split' set to 'True', 'Random seed' set to '123', and 'Stratified split' set to 'False'.

Split Data	
Splitting mode	...
Split Rows	
Fraction of rows in the first output dataset	...
0.8	
Randomized split	...
True	
Random seed	...
123	
Stratified split	...
False	

Figure 12: Split Data.

Step 5: An external testing dataset was used to evaluate model generalisation on unseen data (Ho et al. 2020). The *Id* column was excluded as it contains no predictive value and may introduce noise (*Figure 13*). No missing value treatment was required, indicating high data completeness.

Figure 13: Select Columns in Dataset for Testing dataset.

Step 6: Transformations were learned from the training dataset after data normalisation and consistently applied to the testing dataset to ensure comparability and prevent data leakage, thereby maintaining the integrity of model evaluation (Bouke and Abdullah 2023).

For the Wine dataset, data preparation involves 4 steps.

Step 1, 2, 3 and 4: Selecting all columns, Missing values were handled using median imputation, followed by feature standardisation excluding class target labels, then split data eventually, similarly with the *Step 1, 2, 3 and 4* of Cell_Phone_Price_Prediction dataset.

According to Chaudhry et al. (2023), the class variable was excluded from the train clustering model process as clustering is an unsupervised learning technique that should not use target labels (*Figure 14*). Including this variable would introduce bias and lead to artificially inflated clustering performance. Instead, the class variable is retained only for post hoc evaluation and interpretation of cluster assignments.

Figure 14: Train Clustering Model in Wine dataset.

4. Implementing machine learning methods

4.1 Cell_Phone_Price_Prediction models

Figure 15: Target variable in Train Model.

Two supervised learning models are applied for target variable `price_range` (*Figure 15*): Multiclass or Multinomial Logistic Regression (MLR) and Multiclass Decision Forest (MDF).

Abramovich et al. [2] introduces that MLR is selected as a baseline model due to its simplicity, interpretability, and effectiveness when relationships between features and the target are approximately linear. Given that the dataset consists primarily of numerical variables and has been standardised, MLR is well-suited to capture global linear patterns and produce stable, interpretable results. However, its assumption of linear decision boundaries may limit its ability to model complex feature interactions.

In contrast, MDF used as an ensemble learning method equivalent to Random Forest, capable of capturing nonlinear relationships and complex feature interactions (Talla et al. 2019; Wang et al. 2023). By aggregating multiple decision trees, it reduces variance and improves generalisation, while being less sensitive to feature scaling (Fulazzaky et al. 2024). This makes MDF a complementary model to MLR, enabling a balanced comparison between interpretability and predictive power.

To enhance model performance, hyperparameter tuning is applied. Regularisation (1.0) is used in MLR to control model complexity and ensure stable convergence (Wen et al. 2023; *Figure 16*), while MDF is configured with an increased number of trees (100), controlled depth (10), and minimum leaf size (5) to prevent overfitting and improve stability (Becker et al. 2023; *Figure 17*).



The image shows a configuration interface for 'Multiclass Logistic Regression' in Microsoft Azure. The title 'Multiclass Logistic Regression' is at the top left, with a share icon and a help icon to its right. Below the title, there are five parameter settings, each with a blue information icon and a red asterisk to its left, and a three-dot menu icon to its right. The parameters are: 'Create trainer mode' with a dropdown menu showing 'SingleParameter'; 'Optimization tolerance' with a text input field showing '1e-07'; 'L2 regularization weight' with a text input field showing '1.0'; and 'Random number seed' with a text input field showing '123'.

Parameter	Value
Create trainer mode	SingleParameter
Optimization tolerance	1e-07
L2 regularization weight	1.0
Random number seed	123

Figure 16: MLR's parameters applied in Microsoft Azure.

Multiclass Decision Forest ↗ 📄

Create trainer mode ⓘ * ...

SingleParameter ▾

Number of decision trees ⓘ * ...

100

Maximum depth of the decision trees ⓘ * ...

10

Minimum number of samples per leaf node ⓘ * ...

5

Resampling method ⓘ * ...

Bagging Resampling ▾

Figure 17: MDF's parameters applied in Microsoft Azure.

4.2 Wine models

A single clustering model, K-Means, is applied due to its effectiveness in identifying inherent group structures in numerical data and minimising within-cluster variance. As noted by Ikotun et al. (2023), K-Means is suitable for partitioning data into a predefined number of clusters, with silhouette analysis commonly used to determine the optimal value.

The model is initially configured with 3 centroids to reflect the intrinsic structure of the Wine dataset, ensuring interpretable and well-aligned clusters. According to *Figure 18*, K-Means++ initialisation is used to improve convergence stability, while Euclidean distance is selected as the similarity metric (Coates et al. 2012). Feature normalisation is applied to ensure fair distance calculations across variables with different scales (Coates et al. 2012), and a fixed random seed (123) ensures reproducibility.

To strengthen the analysis, the number of centroids is also set to 4 for comparison. This allows evaluation under both optimal (K=3) and over-segmented (K=4) scenarios, demonstrating that K=3 produces more meaningful, stable, and well-separated clusters that better reflect the natural data structure.

K-Means Clustering

Create trainer mode ⓘ * ...

SingleParameter ▾

Number of centroids ⓘ * ...

3

Initialization ⓘ * ...

K-Means++ ▾

Random number seed ⓘ ...

123

Metric ⓘ * ...

Euclidean ▾

Normalize features ⓘ * ...

True ▾

Iterations ⓘ * ...

100

Assign label mode ⓘ * ...

Ignore label column ▾

Figure 18: K-Means Clustering's parameters applied in Microsoft Azure.

5. Implementing the models in Azure Machine Learning

For the Cell_Phone_Price_Prediction workflow in Azure, the applied pipeline below (*Figure 19*).

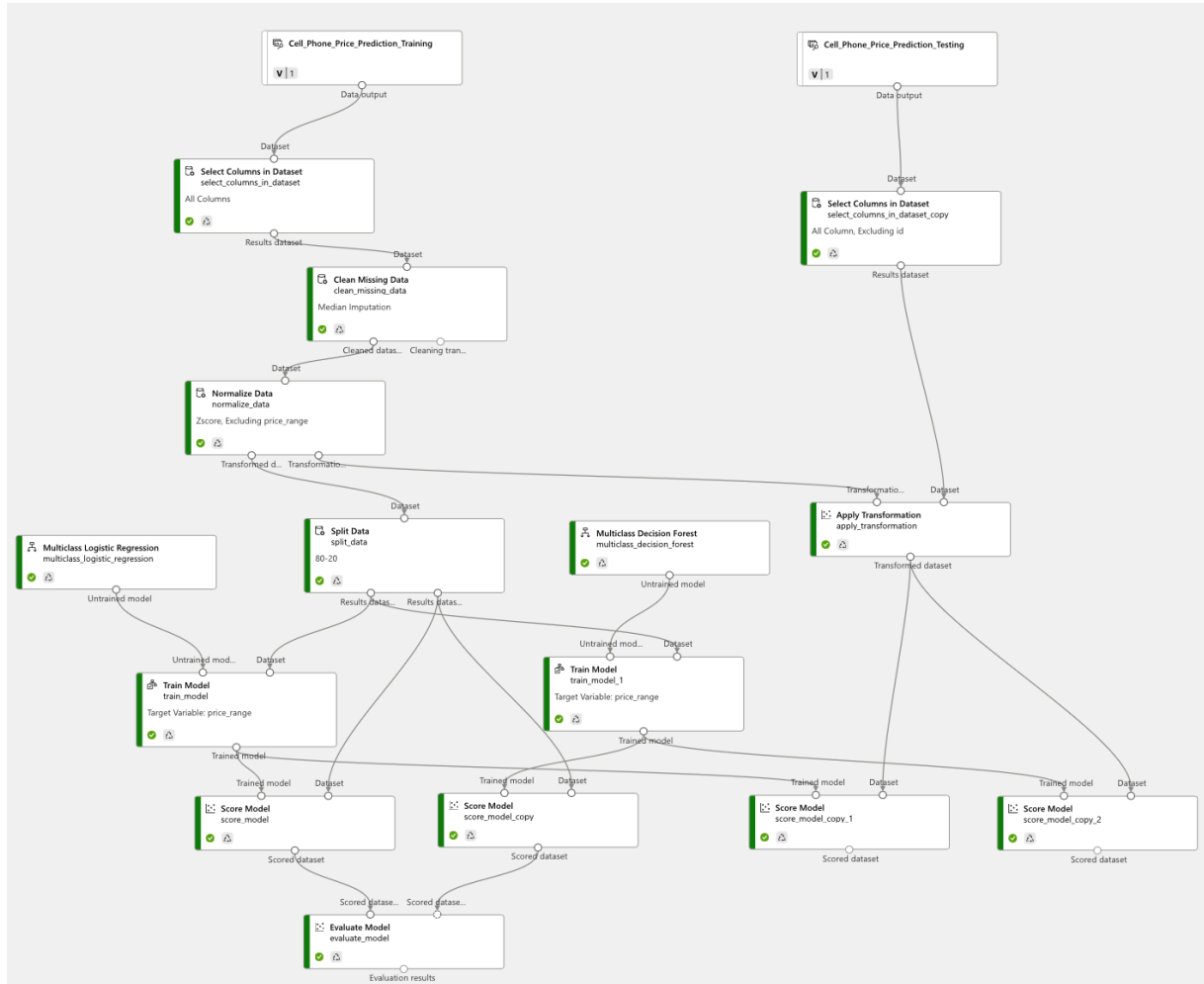


Figure 19: Successfully configured pipeline for Cell_Phone_Price_Prediction datasets in Azure.

For the wine workflow in Azure, the applied pipeline below (*Figure 20*).

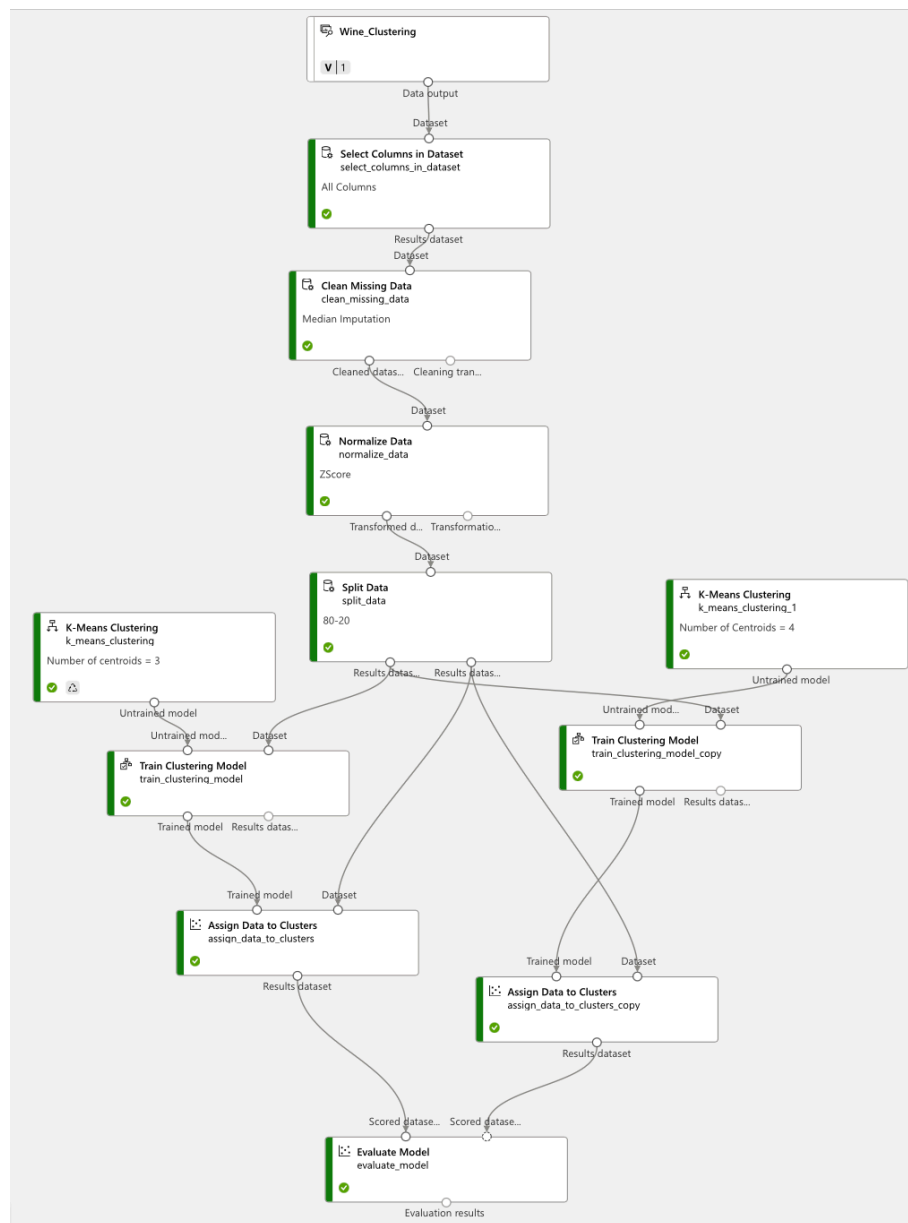


Figure 20: Successfully configured pipeline for Wine datasets in Azure.

6. Analysis of model results

6.1 Cell_Phone_Price_Prediction results

For the cell phone dataset, the key evaluation metrics are accuracy, precision, and recall.

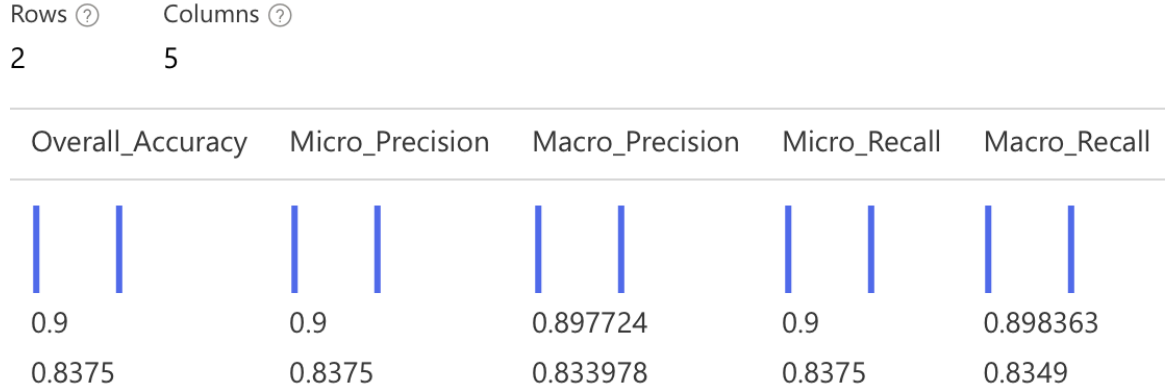


Figure 21: Evaluation result for MLR (Row Above) and MDF (Row Below).

Evaluation metrics such as accuracy, precision, and recall are derived from the confusion matrix, which summarises the relationship between predicted and actual labels. Although only aggregated metrics are displayed, these values are internally computed from the confusion matrix (Riehl et al. 2023).

According to *Figure 21*, MLR consistently outperforms MDF across all metrics, achieving an accuracy of 0.90 compared to 0.8375. It also records higher micro and macro precision and recall, indicating stronger predictive performance and more balanced classification across price categories. This suggests that MLR generalises more effectively and reduces class-specific bias.

The superior performance of MLR indicates that the relationship between technical specifications and *price_range* is largely linear. As a linear model, MLR effectively captures these relationships through weighted feature combinations after normalisation, producing stable and consistent predictions (Nieus et al. 2022).

In contrast, MDF's lower performance suggests that complex non-linear interactions are less critical in this dataset or may lead to overfitting. Despite its ability to model complexity, this does not translate into improved performance (Vallée et al. 2023). Therefore, MLR is selected as the best-fit model, offering higher accuracy, consistency, and practical interpretability.

6.1.1. Final validation

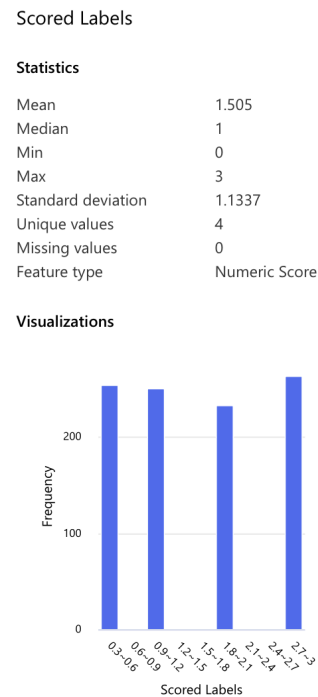


Figure 22: MDF's score label of trained model transformed into testing dataset.

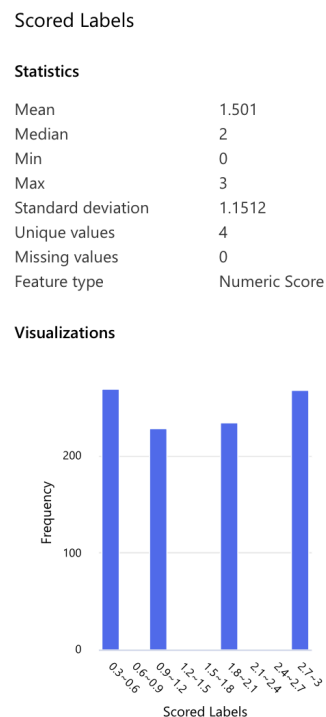


Figure 23: MLR's score label of trained model transformed into testing dataset.

For MDF, the scored labels show a relatively balanced distribution across all price categories, with a mean of 1.505 and median of 1 (*Figure 22*). The standard deviation of 1.1337 indicates

moderate variability, suggesting the model can differentiate across classes without collapsing predictions into a narrow range.

In contrast, MLR produces a distribution with a mean of 1.501 and median of 2 (*Figure 23*), indicating a slight shift towards higher price categories. Its slightly higher standard deviation (1.1512) suggests more dispersed and less balanced predictions (*Figure 23*).

However, despite MDF's more balanced output, MLR achieves significantly higher accuracy and more consistent precision and recall (Owusu-Adjei et al. 2023). This indicates stronger alignment with true class labels, while MDF's balanced distribution does not translate into higher predictive correctness.

6.1.2. Analysis insight

Overall, the results confirm that MLR provides more reliable performance on unseen data. Although MDF produces a more balanced prediction distribution, this does not lead to higher accuracy or better alignment with true labels. In contrast, MLR achieves superior accuracy, precision, and recall, indicating stronger generalisation and more consistent class discrimination. This aligns with the dataset's characteristics, which exhibit largely linear relationships, particularly the strong correlation between RAM and price (~ 0.92), making MLR more effective in capturing underlying patterns and delivering accurate predictions.

6.2 Wine results

Result Description	Average Distance to Other Center	Average Distance to Cluster Center	Number of Points	Maximal Distance to Cluster Center
				
Evaluation For Cluster No.0	0.885103	0.504349	11	0.902684
Evaluation For Cluster No.1	0.996095	0.458936	9	0.563286
Evaluation For Cluster No.2	0.878297	0.618596	16	0.967435
Combined Evaluation	0.909826	0.543772	36	0.967435
Evaluation For Cluster No.0	0.882974	0.504349	11	0.902684
Evaluation For Cluster No.1	0.634079	0.428139	5	0.604601
Evaluation For Cluster No.2	0.859856	0.61961	16	0.977755
Evaluation For Cluster No.3	0.570986	0.412592	4	0.47278
Combined Evaluation	0.803465	0.534796	36	0.977755

Figure 24: Evaluation result for K-mean Clustering with K=3 (the first 4 rows) and K=4 (the rest rows below).

Figure 24 presents the evaluation results for K-Means clustering with K=3 and K=4. The results indicate that the model with K=3 provides superior clustering performance compared to K=4. Specifically, K=3 achieves a higher average distance to other cluster centres (0.9098 against 0.8035), indicating stronger separation between clusters. At the same time, the average distance to cluster centre remains comparable between the two models (~0.54), suggesting that both configurations maintain a similar level of cohesion.

From a data perspective, this result is highly consistent with the structure of the wine dataset, where the true labels (class) are naturally divided into three distinct groups. Therefore, setting K=3 allows the model to align closely with the inherent data structure, resulting in more meaningful and well-defined clusters. In contrast, increasing K to 4 forces the model to artificially split existing groups, leading to weaker separation and less stable clustering patterns.

6.2.1. Final validation

class	Assignments	DistancesToClusterCenter no.0	DistancesToClusterCenter no.1	DistancesToClusterCenter no.2
				
2	2	0.927034	1.210111	0.679375
3	1	1.237171	0.479369	1.1655
1	0	0.276987	1.105584	0.850195
2	2	1.007848	0.969108	0.429934
1	0	0.387518	0.998357	0.630635
1	0	0.354188	1.258103	0.92292
1	0	0.382322	1.18176	0.86897
3	1	1.160343	0.453708	1.040428
1	0	0.42237	1.057314	0.871782
2	2	0.990483	0.849591	0.792687
2	2	1.105722	1.071049	0.967435
1	0	0.665945	1.292753	1.118774
2	2	0.826552	1.116508	0.581566
3	1	1.052691	0.401669	0.932032
2	0	0.902684	1.479884	1.071301
2	2	0.843254	0.894747	0.392378
3	1	1.118308	0.425294	1.03833
2	2	1.141793	0.724207	0.622739
2	2	0.845386	0.89454	0.297567
2	2	1.195043	0.92719	0.697102
2	2	0.751576	1.210347	0.594657
3	1	1.054208	0.371843	0.853748
2	0	0.675716	1.268238	0.724117
2	2	1.061086	1.318553	0.932874
2	2	0.772157	0.894408	0.410441

Figure 25: Results for K=3 in Assign Data to Cluster in Azure.

class	Assignments	DistancesToClusterCenter no.0	DistancesToClusterCenter no.1	DistancesToClusterCenter no.2	DistancesToClusterCenter no.3
					
2	2	0.927034	1.166655	0.680791	1.289391
3	3	1.237171	0.591555	1.172534	0.47278
1	0	0.276987	1.155213	0.849636	1.087006
2	2	1.007848	0.976971	0.426432	1.00612
1	0	0.387518	1.006004	0.628247	1.038258
1	0	0.354188	1.324653	0.92024	1.209945
1	0	0.382322	1.22927	0.869111	1.158688
3	3	1.160343	0.561803	1.04493	0.464902
1	0	0.42237	1.117513	0.869936	1.032641
2	2	0.990483	0.82951	0.791637	0.942578
2	2	1.105722	1.023877	0.977755	1.15687
1	0	0.665945	1.330112	1.120566	1.280969
2	2	0.826552	1.173059	0.578068	1.081605
3	3	1.052691	0.57001	0.936783	0.309766
2	0	0.902684	1.534189	1.068385	1.438142
2	2	0.843254	0.901928	0.396888	0.931047
3	3	1.118308	0.560577	1.042858	0.402919
2	2	1.141793	0.66801	0.625679	0.855834
2	2	0.845386	0.882826	0.297777	0.95592
2	2	1.195043	0.810879	0.702697	1.096289
2	2	0.751576	1.236546	0.59252	1.206862
3	1	1.054208	0.349459	0.859861	0.538491
2	0	0.675716	1.319791	0.719762	1.234005
2	2	1.061086	1.315337	0.93904	1.342117
2	2	0.772157	0.882941	0.415881	0.952294

Figure 26: Result for K=4 in Assign Data to Cluster in Azure.

The clustering results for K=3 provide strong validation of model effectiveness. The Assign Data to Clusters output shows a clear alignment between cluster assignments and true class labels, where each cluster predominantly corresponds to a single class with minimal overlap (Figure 25). This indicates well-defined and compact clusters. Distance measures further support this, with smaller intra-cluster distances and larger inter-cluster distances, confirming strong cohesion and separation.

In contrast, K=4 shows weaker alignment, as observations from the same class are split across multiple clusters—for example, Class 2 across Clusters 2 and 0, and Class 3 across Clusters 3 and 1 (Figure 26). This fragmentation indicates over-segmentation, reduces interpretability, and introduces small, unstable clusters with limited practical value (Figure 26).

6.1.2. Analysis insight

The evaluation results and final validation confirm that K-Means with K=3 is the optimal clustering model for the wine dataset. It demonstrates strong cluster separation and cohesion, with cluster assignments closely aligned to true class labels, indicating high cluster purity and meaningful segmentation. In contrast, K=4 results in over-segmentation, where observations from the same class are split across multiple clusters, reducing interpretability and stability. From a business perspective, K-Means with an appropriate number of centroids effectively

supports wine categorisation for differentiation and decision-making. This outcome is also consistent with the dataset's structure, which naturally forms three distinct groups, showing that K-Means can successfully recover underlying patterns without label information. Therefore, K=3 provides a robust, interpretable, and practically relevant clustering solution.

7. Conclusion

This report demonstrates how machine learning can effectively address different business problems by aligning model choice with data characteristics and objectives. For the cell phone dataset, the analysis shows that MLR performs best, achieving higher accuracy and more consistent predictions than MDF. This indicates that phone price categories are largely driven by stable relationships between technical features such as RAM and battery capacity. As a result, MLR is recommended to support pricing decisions, product positioning, and portfolio planning. For the wine dataset, K-Means clustering with $K = 3$ is identified as the optimal solution, producing clear and well-separated clusters that align closely with the natural grouping of the data and enable meaningful segmentation.

Importantly, Azure Machine Learning played a key role in this analysis by enabling an end-to-end, visual workflow for data preparation, model building, and evaluation. This not only improved efficiency but also ensured transparency and reproducibility of results. Overall, the findings highlight that with appropriate tools and model selection, machine learning can deliver reliable and actionable insights for business decision-making.

8. Reference

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